Assignment 3

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1. Introduction

For this assignment, our goal was to extend our previous work on analyzing Reddit posts about the Ukraine-Russia conflict sentiment pre and post Trump election. Building on the labeled dataset from Assignment 2, our objective in Assignment 3 is to stance classification using both Machine Learning(ML) and Large Language Model(LLM). The goal is to evaluate how well these approaches can identify whether a post expresses a stance that is favorable, opposed, neutral, or irrelevant.

For part 1 of this task, we employed ML techniques to classify the stance of Reddit posts. Using a manually labeled subset of data, we extracted features from text, trained multiple models, and evaluated their ability to predict stance categories such as Favor, Oppose, Neutral and Irrelevant. The goal was to build a model that could generalize well to unseen posts and support large scale automated coding.

For part 2, we used LLM to classify the stance of Reddit posts without applying traditional training. By applying zero-shot and few-shot prompting, we instructed the model to assign stance labels based on natural language understanding. This approach allowed us to compare the performance of LLMs with the Machine Learning models in part1. GitHub Code

Results from Assignment 2

Let's review the result of our assignment 2, our results show that a significant portion of the analyzed Reddit threads were labeled as "Irrelevant." However, a notable shift in stance is observed following the Zelensky's visit to the White House. Posts published after the visit show an increase in opposing views towards the US compared to those shared beforehand. When broken down by subreddit, a similar pattern emerges: many posts reflect a stance that is critical of US involvement. The Ukraine subreddit appears to diverge from this trend, showing that more favorable sentiment toward the US. This contrast may come from the subreddit's focus on Ukrainian perspectives, which could influence the tone of the discourse.

2. Methodology

a. Machine Learning

For the stance detection, we have went with two ML models: SVM and Decision Tree, we trained our model on the 400 hand-coded data points and then we have

1. **Feature Extraction:** For the feature extraction we focused more on converting the words to unigrams and used word2vec also removed punctuation and stop words. We didn't want to remove rarely used words as out project deals a lot in Proper Nouns and

while doing that it was removing those and changing the context of the sentence as a whole, names like ‘Trump’, ‘Biden’ etc may be rare in the corpus but carry significant sway in the swing of stance for our research question.

a. For the word2vec: We went on to use the training type of ‘cbow’ instead of ‘skip gram’’ as CBOW predicts the target word (the word to be embedded) from the surrounding context (a window of words), while Skip-gram does the opposite; it predicts the context (surrounding words) from a given word. CBOW is generally faster and works well when the corpus is large and the vocabulary is relatively dense. It is often preferred when the goal is to learn good representations of words that are contextually similar to each other, as is the case in stance detection, where semantic similarity plays an important role. We also used dim=100 as it's a good standpoint for capturing vocabulary complexity while maintaining computational efficiency.

2. **Dataset Split:** We have split the dataset into 80/20 ratio, 80 for training while 20 for testing. We have gone with the standard train test split ratio.
3. **Model Development:** For the model development, we went on to use SVM (Support Vector Machine) and Decision Tree. The decision to choose SVM was its ability to work well in high dimensional space and in turn in textual Context, the reasoning behind choosing Decision Tree are simpler models that work by splitting the data based on feature values and are interpretable. For SVM we have used the “radial” kernel
4. **Model Evaluation:** We Evaluated the model by constructing Confusion matrix for the models. And analysing the accuracy precision, specificity and sensitivity for each of the models

b. Large Language Models

1. **Model Selection**

For the stance detection task using Large Language Models (LLMs), we selected the model google/gemma-1.1-7b-it from the Hugging Face repository. This model supports instruction tuning and performs competitively in few-shot classification tasks involving political and social context. Our decision to use this model was based on a balance between accuracy and computational efficiency. It offers strong contextual comprehension while being manageable for inference in a university HPC environment.

The choice of using LLMs over traditional ML models was driven by their capability to understand nuanced language patterns and implied sentiment without needing extensive preprocessing or feature engineering.

2. **Prompt Design**

We experimented with both zero-shot and few-shot prompting, but found that few-shot prompting yielded better consistency in stance labeling. For the few-shot setting, we crafted a prompt with 4 labeled examples representing each stance category: Favor, Oppose, Neutral, and Irrelevant. These examples were concise Reddit-style texts that reflected varying attitudes toward U.S. involvement in Ukraine. **LLM Evaluation:** To evaluate the LLM's

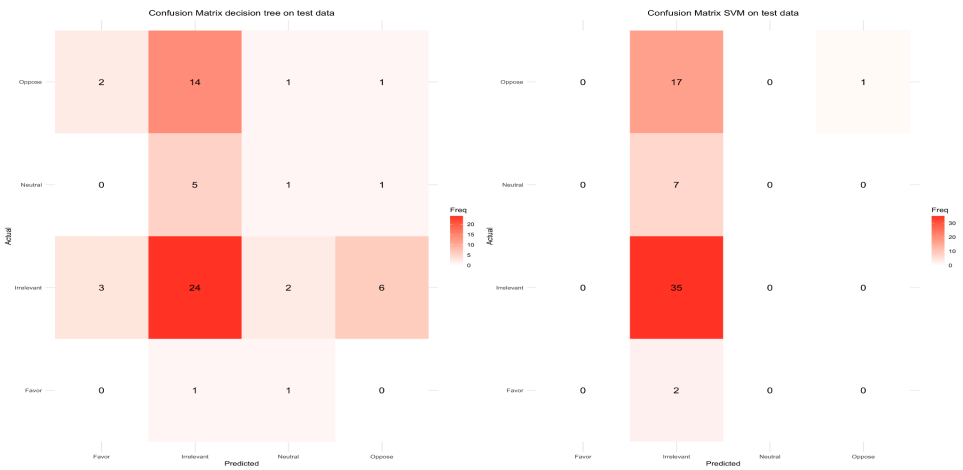
performance, we compared its stance predictions on the test set (20% of the 400 hand-coded dataset) to the original human labels. Metrics such as **accuracy, precision, recall, and F1-score** were computed for each stance category. These were then compared against the performance of the SVM and Decision Tree models from Part 1.

3. Results

a. Machine Learning

We evaluated both the Support Vector Machine (SVM) and Decision Tree models on a 20% test split from the manually labeled data. The SVM achieved higher accuracy (58%) and a kappa of -0.05, showing better overall performance, especially in identifying the dominant *Irrelevant* class(sensitivity 94.3%). However, it still struggled with lower sensitivity on minority classes like *Favor* and *Oppose*. The Decision Tree performed slightly worse (46.77% accuracy, -0.03 kappa) and was also biased toward high-frequency classes.

The following confusion matrices in the figure provide a visual comparison of the prediction performance for the Decision Tree and SVM models on the test dataset. Both models show a strong bias toward predicting the Irrelevant class, which is the most frequent category in the training dataset. In the Decision Tree model, although it correctly predicted a few Oppose and Neutral instances, a large number of posts from those categories were misclassified as Irrelevant. It also produced minor predictions across all classes, but most actual Irrelevant posts were correctly identified. In contrast, the SVM model shows an even stronger bias, it predicted nearly every post as Irrelevant, indicating all actual Oppose, and Favor instances. While this led to high recall for the Irrelevant class, the model failed to identify any post from the other categories.



The structure of the trained Decision Tree helps visualize this behavior. Most internal nodes split based on specific Word2Vec embedding dimensions, but nearly all leaf nodes converge on the *Irrelevant* class. Branches predicting *Oppose* or *Neutral* are sparse and shallow, while *Favor* was entirely unused in

predictions. This further emphasizes the difficulty of separating nuanced stance categories in a low-sample, imbalanced setting.

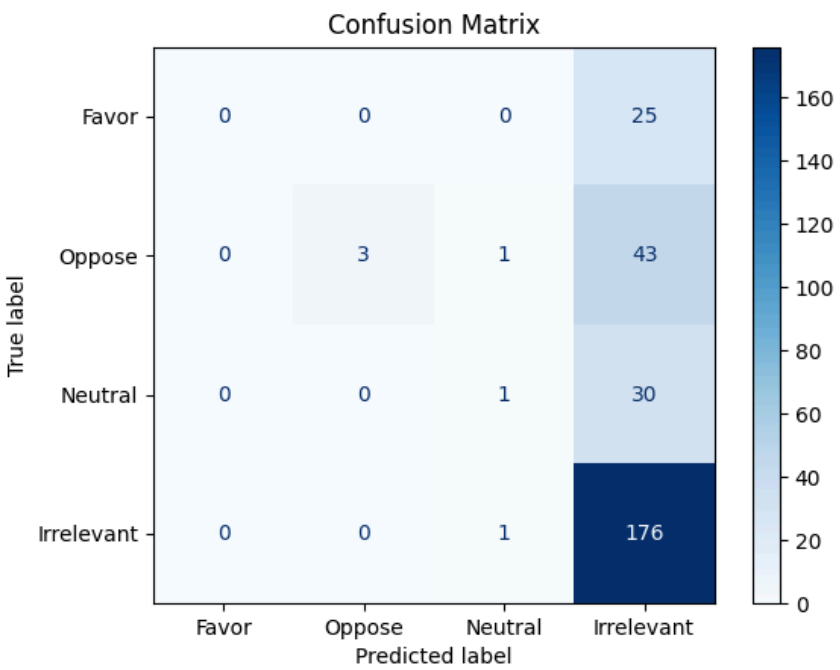
b.Large Language Models

In addition to the traditional machine learning models, we evaluated a large language model (LLM) using Groq's gemma2-9b-it in a few-shot setting. The model was prompted with the question context and asked to choose one of four stance categories: *Approve*, *Oppose*, *Neutral*, or *Irrelevant*.

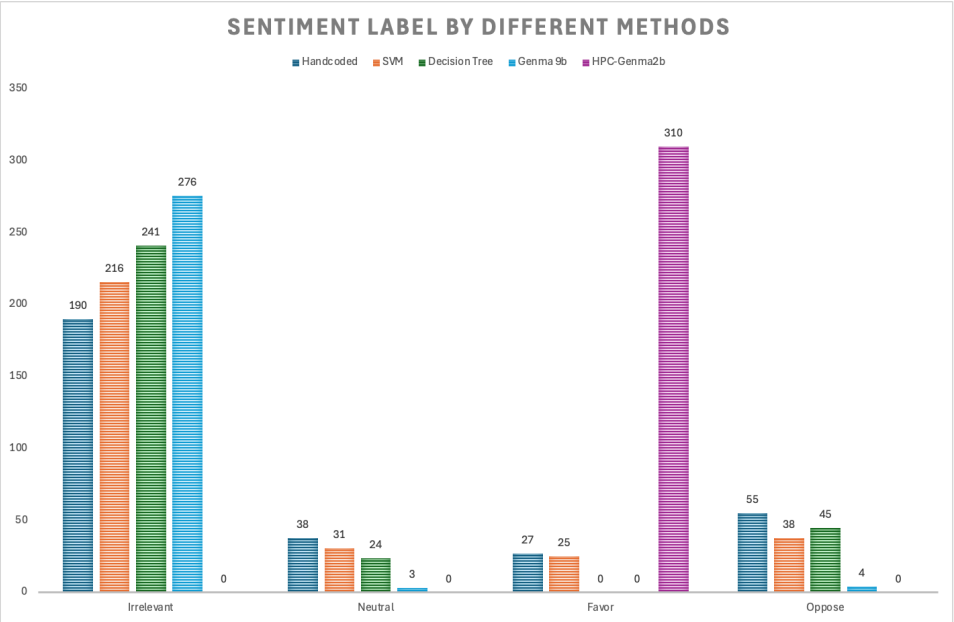
To ensure a fair comparison with the SVM and Decision Tree models, we reproduced the same train/test split used in R (set.seed(123)), and evaluated the LLM predictions on the corresponding 20% test set. This controlled for data leakage and enabled a direct performance comparison across all three methods.

The LLM achieved 56.5% accuracy, but its precision and recall were extremely low (19% and 7.5%, respectively). Warnings during evaluation indicated that some classes were never predicted, meaning the model heavily favored certain labels while ignoring others. For example, it tended to default to a single class, resulting in very poor coverage of minority classes like *Oppose* or *Favor*.

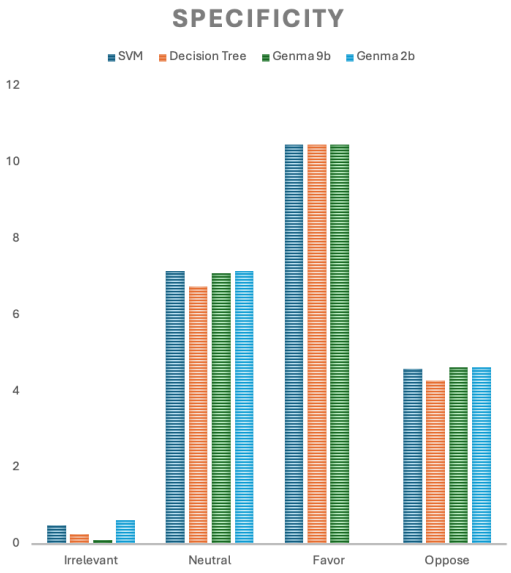
These results suggest that zero-shot prompting without examples may not be effective for this kind of imbalanced, fine-grained stance detection task. While the LLM is capable of generating responses, its performance in structured classification remains limited in this context.

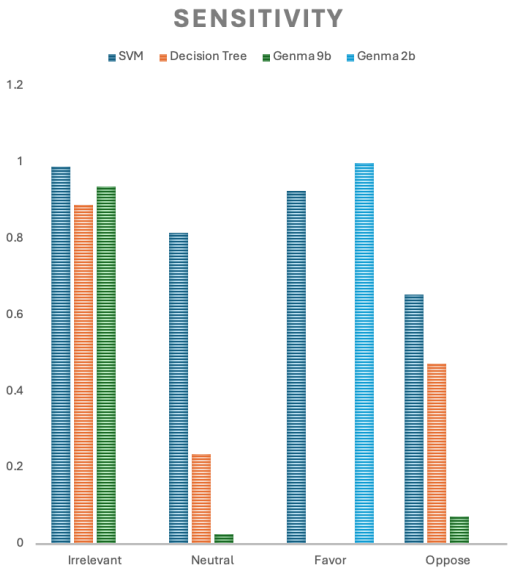


4. Discussion



We noticed that SVM was working the best among all i.e LLM's and ML algorithms, the thing which was of note was due to the class imbalance accuracy was not the only deciding factor for the best model predictor, since we have huge class imbalance of irrelevant data point in our dataset, for instance we achieved an accuracy of 56% when we changed the 'cp' parameter for the decision tree but upon closer inspection we noticed that this model was labeling everything as irrelevant, so we decided to discard this model and went with the much conservative decision tree of 46% accuracy from before. We also notice a similar distribution to hand coded sentiment analysis being reflected much more closely in SVM than any other model.





If we look at the specificity chart we can note that most of the models have struggled predicting favor genma 2b was the best but as we know it was predicting mostly every thing to favor. This was due to the fact that may be the most though having a title have been described in the image like suppose we have a post with a title “Check this out” with an image attached which may have been of oppose or favor but without this context we labelled them as irrelevant. One thing we could have done was synthetically increasing the data points for other labels or would have used stratified sampling, but since our training data was small from the start we didn't have this luxury.

5. Conclusion

Our analysis shows that the Support Vector Machine (SVM) model performed best overall in the stance classification task. It achieved the highest accuracy and sensitivity, particularly in identifying the dominant Irrelevant class. However, like the Decision Tree and LLM models, it still struggled with minority stance categories such as Favor and Oppose, likely due to class imbalance in the data.

The Decision Tree model was more limited, showing a stronger bias toward high-frequency classes and failing to predict certain categories altogether. Its structure confirmed this tendency, with most branches converging on Irrelevant and shallow splits for other classes.

The LLM model, evaluated using the same test split, showed limited effectiveness. Although it achieved similar accuracy to the SVM on the test set, its precision and recall were very low, and warnings during evaluation indicated that several stance categories were not predicted at all. This highlights the challenges of applying zero- or few-shot prompting to imbalanced, fine-grained classification tasks without further tuning or adaptation.

Overall, our results suggest that while LLMs offer flexibility, traditional machine learning models—especially SVMs—still provide more reliable performance in this specific stance detection context, particularly when the dataset is small and imbalanced.